

Tuning and Profiling LAMMPS on Deucalion

Large Scale Hybrid Computing 25/26

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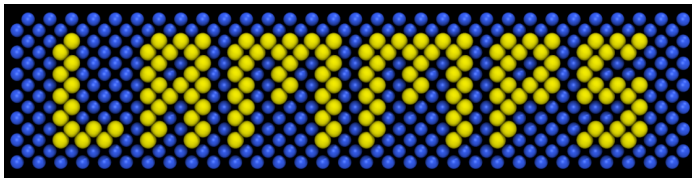
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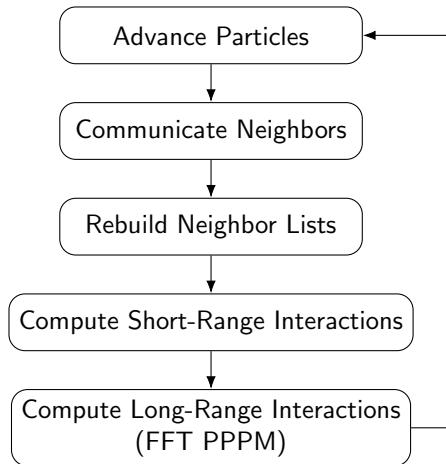


- Scientific simulations require ever more computation power;
- HPC systems offer substantial performance potential;
- LAMMPS is widely used for performing Molecular Dynamics simulations.



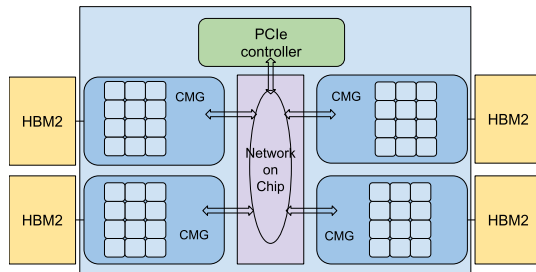
LAMMPS (Sandia Corporation)

LAMMPS



Deucalion ARM Partition

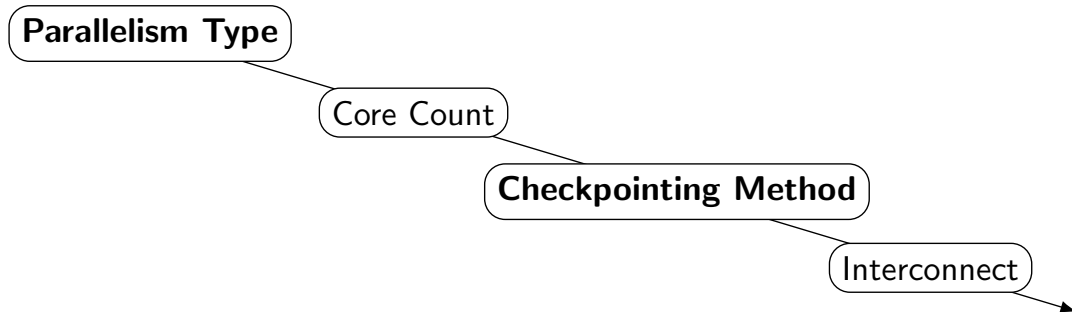
- 48-core Fujitsu A64FX with NUMA;

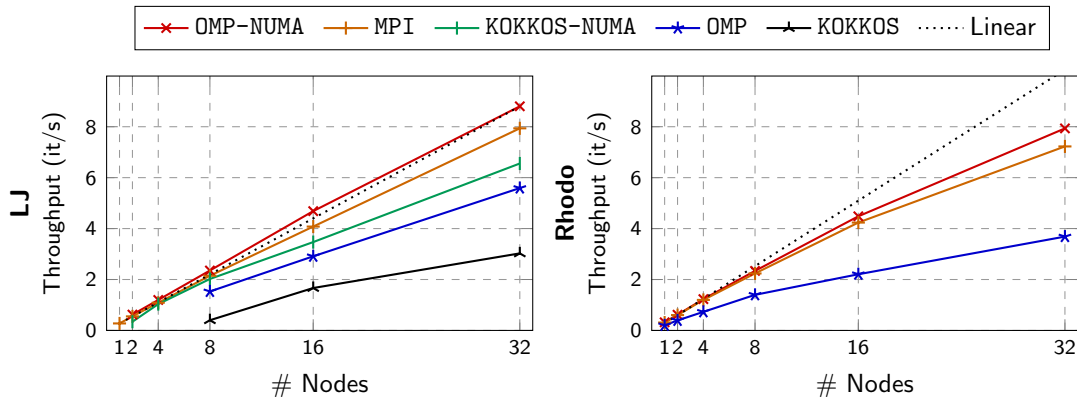


- Infiniband and Ethernet interconnects;
- LustreFS Parallel File System (PFS).



- Benchmarks with distinct computation characteristics:
 - **LJ**: short-range interactions only, compute-bound;
 - **Rhodo**: short and long range interactions, communication-sensitive.
- Cascading optimization with scalability analyses:



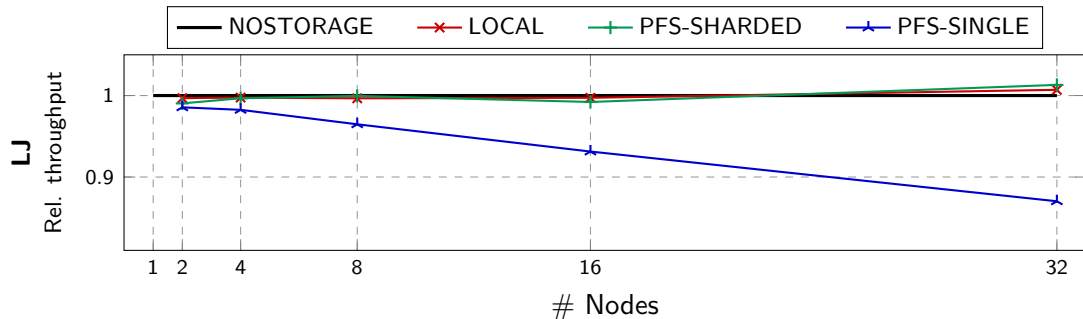


- Linear scaling in LJ, but not Rhodo;
- Consistently better performance with OMP-NUMA (up to 10.8% gains).

Tuning – Checkpointing Strategy



- Full simulation checkpoint every 100 iterations;

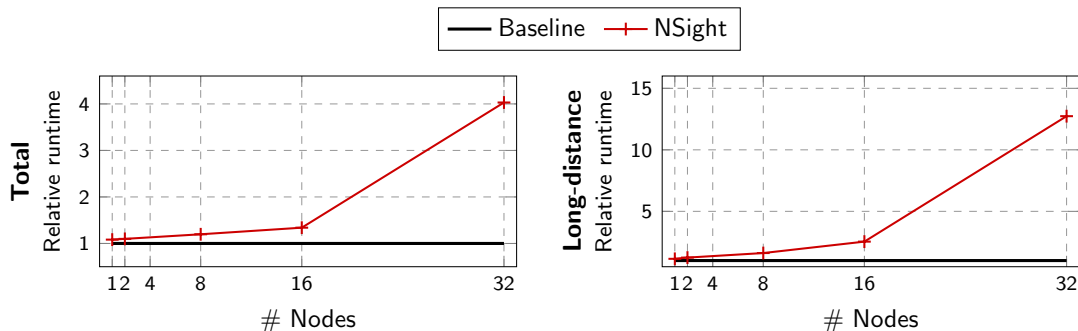


- Checkpointing without significant overhead is possible;
- Up to 13.0% performance loss with PFS-SINGLE (mainly due to stragglers).

Profiling – Methodology and Overhead



- Questions:
 - Kernels with optimization potential (1-node profiling);
 - MPI communication breakdown and load imbalance (1-node vs 32-nodes);
 - ...
- High profiling overhead:





Symbol	Self (%)	Module Name	CPU Time (%)
PairLJCharmmCoulLongOMP::compute	62.51	liblammps.so.0	96.87
NPairBinOmp<...>::build	14.71	libm-2.28.so	1.53
PPPMOMP::fieldforce_ik	7.11	libgomp.so.1.0.0	0.51
PPPMOMP::make_rho	2.39	libfftw3.so.3.6.10	0.44
Others	13.28	Others	0.65

Single node Rhodo call breakdown

Single node Rhodo module breakdown

- Dominant compute-bound kernel → compiler changes, SIMD implementation;
- Little opportunity for optimized external library.



MPI Directive	1-node		8-node	
	MPI Time (%)	Wall Time (%)	MPI Time (%)	Wall Time (%)
MPI.Wait	41.0	1.32	41.0	7.69
MPI.Allreduce	31.0	1.01	28.0	5.30
MPI.Send	21.0	0.68	16.0	2.98
MPI.Waitany	–	–	7.0	1.45
Others	7.0	–	8.0	–

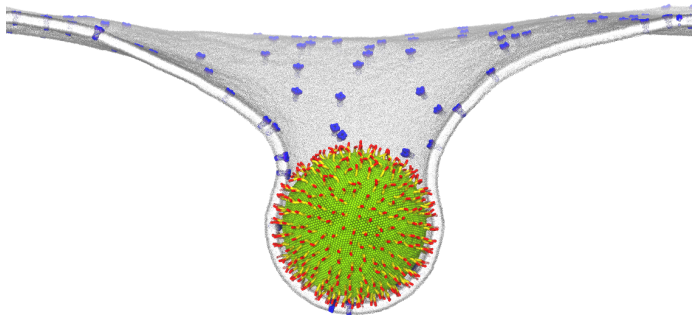
MPI directive time on 1-node and 8-node configurations

- Growth in communication time with node count (3.1% → 17.7%);
- Reported performance disparities of up to **25.1%** between MPI ranks;
- Collective MPI directives, non-blocking MPI directives, ...

Conclusion



- MPI+OpenMP NUMA-aware setup achieved performance improvements of 10.8%;
- Sharding can eliminate the performance overhead of simulation checkpointing;
- Profiling revealed future work opportunities.



LAMMPS Simulation Visualization (Comp Phys Comm)

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